Can ChatGPT Teach Tendon Repair Suturing? A Comparison With Video Instruction in a Simulation-Based Environment: A Pilot Study.

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Journal: Journal of surgical education | **Year:** 2026 | **DOI:** 10.1016/j.jsurg.2025.103867

Link: https://doi.org/10.1016/j.jsurg.2025.103867 (open with institutional access)

Abstract

To compare the effectiveness of ChatGPT-guided instruction versus video-based instruction for teaching the Modified Kessler tendon repair technique to novice medical trainees in a simulation setting. A quasi-experimental, non-equivalent control group, post-test only study was conducted between April and June 2025. The study was conducted at a teaching simulation center in Ecuador, between April and June 2025 PARTICIPANTS: Thirty-eight third-year medical students without prior surgical experience were recruited. After standardized training in basic suturing, 33 students who demonstrated competency in the simple interrupted suture objective structured assessment of technical skills (OSATS ≥ 26/40) were included. Participants were assigned to 1 of 2 self-directed groups: ChatGPT-Tutored (n = 17) or Video-Tutored (n = 16). The ChatGPT group received real-time, stepwise guidance via a predefined instructional prompt with conversational feedback, whereas the Video group watched a structured video showing each procedural step. Both groups practiced for 5 hours on a tendon simulator. The Video-Tutored group achieved significantly higher tendon-specific OSATS scores (median 30 interquartile range [IQR 24-32.75]) compared with the ChatGPT-Tutored group (median 18 [IQR 15.25-22.75]) (p = 0.0004). Competency (≥30 points) was reached by 52.6% of Video-Tutored students versus 10.5% of ChatGPT-Tutored students (p = 0.002). No participant in either group achieved mastery (≥38 points). Satisfaction (median 22 [21-23.75] vs. 8 [7-9.75]; p < 0.0001) and self-confidence (mean 35.26 ± 2.62 vs. 10.21 ± 3.19; p < 0.0001), measured with the NLN student satisfaction and self-confidence in learning scale, were also significantly higher in the video group. Video-based instruction proved more effective than ChatGPT guidance for teaching tendon repair suturing to novice trainees, producing superior technical performance, satisfaction, and confidence. Nonetheless, conversational AI remains a promising adjunct where tutors are unavailable. Future studies should test blended video-AI approaches to optimize surgical education.

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